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TEM study of the geoelectrical structure and groundwater salinity of the Nahal Hever sinkhole site, Dead Sea shore, Israel

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ABSTRACT

Since the 1990s a large number of sinkholes have appeared in the Dead Sea (DS) coastal area. Sinkhole development was triggered by the lowering of the DS level. In the literature the relationship between the sinkholes and the DS level is explained by intrusion of relatively fresh water into the aquifer thereby dramatically accelerating the salt dissolution with creation of subsurface caverns, which in turn cause sinkholes. The main goal of our project was detection and localization of relatively fresh groundwater. During our study we used the transient electromagnetic method (TEM) to measure the electrical resistivity of the subsurface. As a test site we selected Nahal Hever South which is typical for the DS coast. Our results show that resistivity of the shallow subsurface reflects its vertical and lateral structure, e.g., its main hydrogeological elements explain the inter-relations between geology, hydrogeology, and sinkholes. The TEM method has allowed detailed differentiation of layers (clay, salt, etc.) in the subsurface based on their bulk resistivity. The 10 m-thick salt layer composed of idiomorphic crystals of halite deposited during the earlier Holocene period was extrapolated from borehole HS-2 through the study area. It was found that in Nahal Hever the typical value of the bulk resistivity of clay saturated with the DS brine varies between 0.2 and 0.3 Ωm, whereas saturated gravel and sandy sediments are characterized by resistivity between 0.4 and 0.6 Ωm. The high water salinity of the aquifer (enveloping the salt layer) expressed in terms of resistivity is also an important characterization of the sinkhole development mechanism. The electrical resistivity of the aquifer in the vicinity of the salt unit and its western border did not exceed 1 Ωm (in most cases aquifer resistivity was 0.2–0.6 Ωm) proving that, in accordance with existing criteria, the pores of the alluvial sediments are filled with highly mineralized DS brine. However, we suggest that the criterion of the aquifer resistivity responsible for the salt dissolution should be decreased from 1 Ωm to 0.6 Ωm corresponding to the chloride concentration of approximately 100 g/l (the chloride saturation condition reaches 224 g/l in the northern DS basin and 280 g/l in the southern one).
Based on TEM results we can reliably conclude that in 2005, when most of sinkholes had appeared at the surface, salt was located within a very low resistivity environment inside sediments saturated with DS brine. Intrusion of relatively fresh groundwater into the aquifer through the $600 \times 600\ m^2$ area affected by sinkholes has not been observed.

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1. Introduction

Since the 1990s thousands of sinkholes have appeared in alluvial fans and other unconsolidated sediments along the coastline of the Dead Sea (DS) in Israel and Jordan. Sinkholes pose the main threat to economic development of the western (Israel) and eastern (Jordan) shores of the DS area.

The development of sinkholes is explained in the literature by rapid dissolution of buried salt with the concomitant creation of large caverns (Frumkin and Raz, 2001; Yechieli et al., 2002). These salt layers, composed of idiomorphic crystals of halite, are estimated to be 10,000 years old. They are water-saturated and very conductive hydraulically. The halite layers were deposited during the period between the Pleistocene and the Holocene owing to the development of extremely arid conditions in the shrinking phase between the Lisan Lake and the present Dead Sea (Yechieli et al., 1995).

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The sinkhole development is triggered by the lowering of the DS level that is accompanied by a corresponding lowering of the groundwater level. Lowering of the DS level allows the incursion of relatively fresh groundwater into the coastal aquifer (Yechieli et al., 2002). Indeed, when the DS level was high, a salt formation typically located at a depth of 25–50 m (top) was surrounded by very saline groundwater similar in composition to the DS water, with total dissolved solids [TDS] of 340 g/l. DS water also has very high chloride (Cl^-) concentration (up to 224 g/l), which constitutes 98% of the anions, with little sulfate and carbonate. Chloride concentration is therefore the main parameter used to characterize groundwater (Yechieli, 2000). Migration of fresh or unsaturated (in relation to Cl^-) groundwater into the coastal area accelerates the dissolution process at the zone of contact of groundwater with the salt layer (Frumkin and Raz, 2001; Yechieli et al., 2002).

As explained above, the model based on the rapid dissolution of buried salt requires intrusion of relatively fresh water into the aquifer. To learn more about variations of the salinity of groundwater and its influence on sinkhole development, we performed a geophysical study in the Nahal Hever South (NHS) area, which is typical of the DS coast.

The TEM method is routinely used in Israel for locating the fresh–saline water interface in coastal areas (Goldman et al., 1991; Kafri and Goldman, 2005). Different classifications based on bulk resistivity of aquifers have been suggested to estimate water salinity in sandy–gravel aquifers (Kafri et al., 1997; Yechieli et al., 2001). We will examine this issue in detail in Chapter 4. Note here that the above-mentioned conventional TEM (TDEM) method (Fitterman et al., 1988; Goldman et al., 1991), and especially its FAST technique (Barsukov et al., 2007), are highly efficient instruments for mapping the resistivity structure of aquifers and estimate the salinity of groundwater that saturates the sediment pores. On the other hand, bulk resistivity of aquifers is the function of both the resistivity of water filling the pores and of the porosity of sediments. Porosity is commonly assumed to be constant throughout the investigation area. The only possibility to use different porosities for the interpretation of TEM data is when some additional information about the geological formation (for instance, from boreholes) is available (Keller and Frischknecht, 1966). In addition, the estimation of water salinity using bulk resistivity is complicated in clayey sediments (McNeill, 1980).

2. Geology and hydrogeology of the DS coast

The DS is the terminal lake of the Jordan River system, located 420 m below sea level in an extremely arid environment with an annual precipitation of 50–100 mm (Fig. 1).

Since the early 1960s the DS level has continuously dropped at a rate of about 0.5 m/year in 1960. This rate increased to 0.8 m/year in 1980; recently the rate has further increased to 1.0 m/year (Yechieli et al., 2004). Until the mid-1960s the DS level kept dropping both in the northern basin (DS) and in the evaporation ponds (EPs) located to the south of the DS (see Fig. 1). Both sodium chloride and potassium salts (used worldwide in chemical manufacture) are produced from the DS evaporation ponds. Since 1980 the level of the EPs was increased by artificial pumping of water from the northern basin into the EPs, which are currently maintained approximately 30 m above the DS level (Yechieli, 2000).

A number of observation boreholes have been drilled by the Geological Survey of Israel (GSI) along the Dead Sea coast (Yechieli et al., 2004). These boreholes are located in seven sites affected by sinkholes (Fig. 1) At six of seven sites no fresher water was detected neither at the nearest salt vicinity nor at underlain aquifers. Chloride concentration in groundwater varied in the range of 120–224 g/l or 55–100% of chloride saturation (Legchenko et al., 2009).

The Nahal Hever South (NHS) test site is located in the central part of the western shore of the northern DS basin (Fig. 1). Since 1991, 18 sinkholes have formed in this area; however development of sinkholes is now slowing down (Legchenko et al., 2008a). The subsurface is composed of 15–25 m thick alluvial fan sediments (sand and gravels with clay intercalations), a 1–5 m thick marl layer,

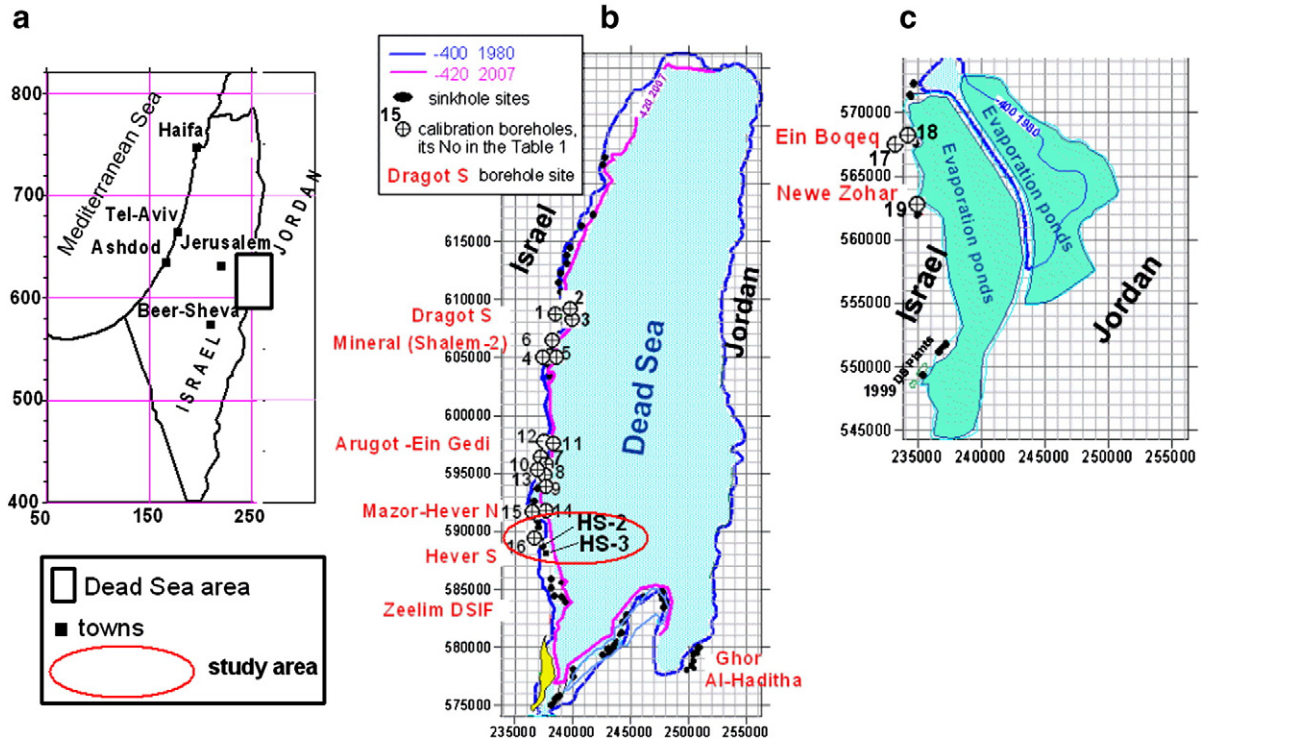


Fig. 1. Location map of the study area (a); and calibration boreholes along the western DS shore: (b) northern basin and (c) southern basin (evaporation ponds). Black circles designate the calibration boreholes numbered as: 1 — Dr2, 2 — Dr3, 3 — Dr4, 4 — Mn1, 5 — Mn2, 6 — Mn4, 7 — EG6, 8 — EG7, 9 — EG11, 10 — EG13, 11 — EG17, 12 — EG18, 13 — EG19, 14 — Mz1, 15 — Mz2, 16 — HS2, 17 — EB1, 18 — EB2, and 19 — NZ1.

and a 1–15 m thick salt layer. A clay and gravel formation underlies the salt layer. Iron oxide mineralization is visible in open pits (Arkin and Gilat, 2000). Geological sections of two boreholes drilled in Nahal Hever (HS-2, HS-3) are presented in Fig. 2 (Yechieli et al., 2004).

Borehole HS-2 detected a solid salt layer at a depth of 24–35 m. In borehole HS-3 (40 m from HS-2) a dense mud was detected at a depth of 24–28 m, which is assumed to be a dissolution cavity in buried salt found in HS-2. The water table in the area was measured in 1999 at a depth of 17.2 m and two years later at 18.5 m. The chloride concentration was 186–212 g/l Cl (i.e., ~83–95% of saturation level).

3. Transient electromagnetic method

The transient electromagnetic method (TEM) also referred in the literature as the time-domain electromagnetic method (TDEM), is sensitive to bulk resistivity (conductivity) of the studied medium. It is a particularly good method for identifying low resistivity targets. Generally, the resistivity of the subsurface depends on several parameters, most important of which is salinity of the fluid in the pores and porosity. Quantitative interpretation of resistivity is based on the modified *Archie's Law* (Archie, 1942), which establishes that in saturated geologic materials with conductive water in the pores the bulk electrical resistivity depends on the porosity and the resistivity of the pore fluid. In this case bulk resistivity is expressed as:

$$\rho_x = a\rho_w\phi^{-m}, \tag{1}$$

where ρ_w is the resistivity of the solution filling the pores, and ϕ is the porosity. Parameters a and m depend on the geometry of the pores. Eq. (1) can be further simplified if we accept $a=0.81$ and $m=2$ for sandy DS aquifers (Kafri and Goldman, 2005):

$$\rho_x = 0.81\rho_w\phi^{-2}. \tag{2}$$

It follows from Eqs. (1) and (2) that there is direct dependence connecting bulk resistivity (ρ_x) measured by the TEM method with a water resistivity (ρ_w). The latter is a function of the water salinity as measured by chloride concentration or total dissolved solids (TDS) (Keller and Frischknecht, 1966). This relationship allows interpretation of bulk resistivity in terms of salinity.

From the above equations it is impossible using only TEM measurements to determine both the water resistivity and porosity (ϕ). Usually, parameters ρ_w and ϕ are not known, and porosity of them is assumed to be constant throughout the investigation area. When this assumption is not true, some additional information about the geological formation (for instance, from boreholes) is required for correct interpretation of ρ_w from the TEM data. Porosity ϕ in Eq. (2) could be also calculated from other geophysical parameters, such as shear wave velocity V_s (Dvorkin, 2008). However, in the present paper we consider water salinity based on the relative criteria suggested by Kafri et al. (1997) and on our resistivity–salinity calibration (Ezersky and Legchenko, 2009).

Previous TEM measurements before 2004 were carried out using an EM-37 GEONICS instrument with the central loop configuration (Goldman et al., 1991; Shtivelman et al., 1999). Since 2005 we have used the TEM FAST 48HPC system with a single (coincident) Rx/Tx loop (AEMR, 2005). Kamenetsky (1997) noted the important role of the TEM-FAST methodology, device, and software in shallow TEM investigations. Because TEM FAST data have not been extensively discussed in the literature, we will take a closer look at this system.

The TEM FAST 48HPC system comprises a single Transmitter–Receiver–Controller unit (<2 kg mass) managed by the HP IPAQ pocket. A single square (rectangular) loop is used both as a transmitter (Tx) and as a receiver (Rx) antennae, so-called single or coincident loop configuration. Antenna is made of wire (0.5 mm^2 for 25×25 m^2 loop in our case) providing high productivity in field conditions. A test ring controls system stability. Time of measurements from 4 μs to 16 ms includes 48 time windows, with repeatability frequency changing from 3.2 kHz to 11 Hz, respectively. The longer the transient time, the greater the distortion of measurement results caused by noise. Usually, the time range can be expanded as long as the measurement errors for a signal at maximum time do not exceed 10–20%. Ramp time (delay in starting the measurements) depends on loop size (usually, it varies from 155 ns for a 10×10 m^2 loop to 1720 ns for a 100×100 m^2 loop). The dynamic range of voltage received (max/min) is between 10 V and 100 nV, which is the self noise of the system. The specifications of the TEM FAST 48 HPC system can be found in the TEM FAST Manual (2005). The principles of the method are described in Barsukov et al. (2007).

In the TEM-FAST methodology coincident loop enables three objectives to be reached at the same time:

(a) Effects of subsurface irregularities in the antenna vicinity are integrated maximally;
(b) Efficiency of the field experiment grows significantly because central receiver loop is no longer necessary. In addition, light

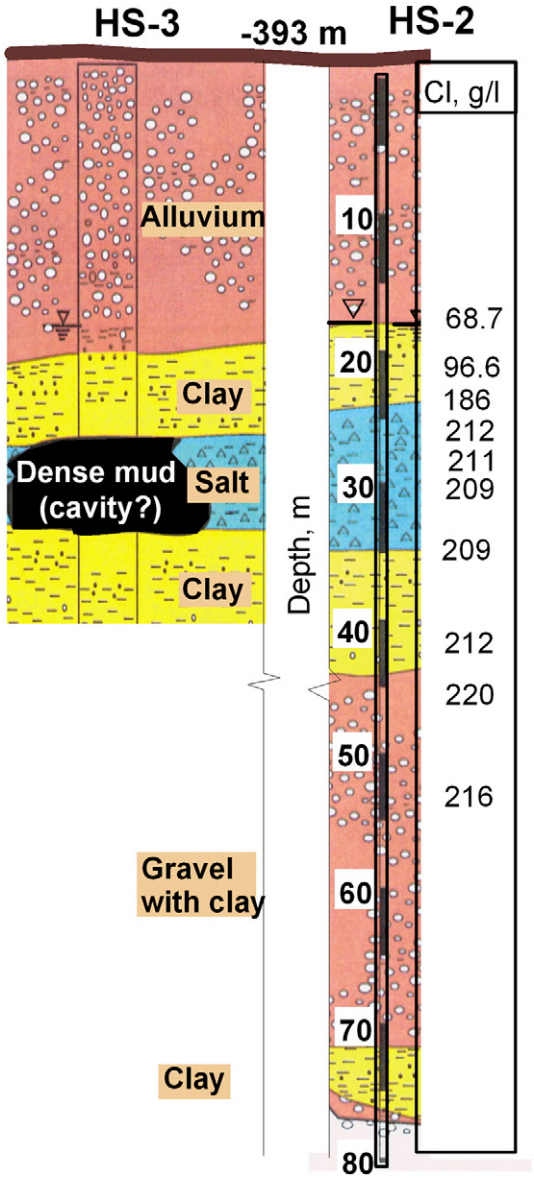


Fig. 2. Geological cross-section based on boreholes HS-2 and HS-3. Water level and the chloride concentration were measured in 1999.

and compact device allows faster data acquisition at less field crew (Albouy et al., 2001);

(c) The data processing is based on performing the transformations allowing both fast data processing in the field conditions and automatic selection of the start model for the data inversion.

On the other hand, TEM results with a coincident loop may be less reliable than those made with a central loop configuration. One of the factors affecting the coincident loop configuration measurements is magnetic viscosity, or the super paramagnetic effect (SPM) (Barsukov and Fainberg, 2001). Additional errors in TEM data processing may be caused by the induced polarization (IP) effect and the antenna polarization (APE) effect (Barsukov et al., 2007; Kaufman et al., 1989; Lee, 1981). These side effects affect both the central loop and the coincident configuration. However, in the second case they are exacerbated, and should be accounted for with the help of the software used to process the data. In this paper we consider the SPM effect.

The SPM effect in rocks is related to orientation/disorientation processes of magnetic moments in very fine grains of the magnetic field. The most intensive SPM effects are observed in regions composed of effusive and volcanogenic-sedimentary rock development. In the central loop configuration, the expression of SPM effects is considerably weaker than in the coincident configuration. Nevertheless, the rocks in the study area are characterized by low magnetic susceptibility ($5–8\times10^{-5}$ SIU, Rybakov et al., 2005), and therefore we can expect a slight influence of the SPM effect. We will compare the results acquired at the same place using both the central and the coincident loop configurations, which are sensitive to the SPM effect in different degrees.

3.1. Comparison of results obtained with the EM37 (Geonics) and TEM FAST (AEMR)

We carried out a comparison of the transient signals and resistivity–depth functions obtained using the Geonics EM-37 with central loop and TEM FAST 48HPC systems with coincided loop configurations in the NHS at approximately the same location. We used data acquired at station 5W with the Geonics EM-37 system in 2001 (Goldman et al., 1991), and data acquired by us at station 2 in 2005 with the TEM FAST 48HPC system (AEMR, Netherlands). The location of both stations is shown in Fig. 6. Fig. 3a and b (left panels) shows transient signals measured at stations 2 and 5W, respectively (see Fig. 6 for location). The right panels of Fig. 3a and b demonstrate the inverse resistivity–depth functions.

Both measurements were processed using the IX1D software (Interpex, 2006). The EM-37 data comprise the 28 ms time range (measurements were carried out up to 70 ms but the data was noisy at late stage repetition frequency was of 2.5 Hz) and the TEM FAST set 16 ms time range (repetition frequency 11 Hz). It can be seen in Fig. 3a and b that the shapes of both transient signals are similar in the same coordinate systems "apparent resistivity–transient time." Moreover, we combined the Geonics and TEM transient curves in Fig. 3b (left) to demonstrate the similarity of the two signals on the same time scale.

Inversion of the two data sets was performed for a 7-layered model. The inverse resistivity versus depth functions of the two soundings presented in the right panels of Fig. 3a and b attests to the fact that two different systems can be characterized by highly similar inversion models. Some discrepancy at the shallow depth range could be related to different loop sizes ($50\times50\ m^2$ for EM-37 and $25\times25\ m^2$ for the TEM FAST). In addition, 3 years passed between the measurements, which can also affect the results. The comparison of results obtained using the EM GEONICS and TEM FAST systems revealed high correspondence, suggesting that (a) super paramagnetic effects (SPM) are negligible under the local conditions, and (b) results acquired using EM-37 systems at earlier stage of the sinkhole development can be used for temporal monitoring with the TEM FAST system.

3.2. TEM-Researcher (AEMR) and IX1D (Interpex) software packages

To obtain reliable interpretations of the TEM measurements performed with the TEM FAST equipment, we used two 1D inversion software packages.

IX1D (Interpex, 2006) is considered to be the standard software for interpreting Geonics TEM data, allowing forward and inverse modeling for layered and smoothed (1D) models, equivalence analysis, and calculation of the root mean square (RMS) misfit error for every iteration. The RMS misfit error is used as an indication of the fit between the synthetic data generated from the model and the measured data. For TEM data (in-loop), RMS is estimated in the log domain. The error is estimated according to the following formula:

$$E_{log} = \sqrt{\left\{\sum_{i=1}^{i=n}\left[\ log(V_s / V_m)\right]^2\right\} / n} \tag{3}$$

where V_s is the synthetic field value (voltage) and V_m is the measured value and n is the number of samples (measurement points). The RMS log error can be converted to linear space and expressed in percents as follows:

$$E\% = \left(10^{E_{log}} - 1\right)*100. \tag{4}$$

Equivalence of the inversion parameters of selected layers is estimated based on the Resolution Matrix, which is a byproduct of the inversion process (Stroyer, 2010). The matrix is used in the equivalence analysis process. When the inverse process reaches the best fit the parameters are incrementally adjusted until the fitting error reaches 1.2 times the best fit error.

The *TEM-RESEARCHER* is intended for modeling and inversion of TEM sounding data acquired with the TEM FAST 48 HPC system (Barsukov et al., 2007; TEM-RESEARCHER, 2005). There are two ways of generating a section: transformation and inversion. The transient curves $\rho_a(t) = \mathrm{Res}(t)$ are transformed into a pseudo-section $\rho(h) = \mathrm{Res}(h)$. The layered model is selected based on the transformation curve, and layered inversion is performed. The software allows processing data for coincident and central loop configurations. It includes an option for taking into account the Induced Polarization (IP) and Super Paramagnetic (SPM) effects. The software generates both the transformation and the inversion sections and maps. It is designed for fast performance of "quasi" 2D and 3D pseudo- and inverse-resistivity sections and maps from multiple 1D soundings (tens–hundreds of soundings), which should be performed often under field conditions. Therefore, the option for estimating the equivalency and misfit error is not included in the software. For more accurate estimations of the inversion results, IX1D by Interpex Ltd. must be used.

The combined use of the two software packages allows taking advantage of the IX1D capabilities, at the same time ensuring that TEM FAST data are correctly inverted. Our experience shows that both software packages provide highly similar results, whether used with Geonics or TEM FAST data. An example of such interpretation of a data set obtained at TEM station 21 is shown in Fig. 4.

The left panels (a) and (c) of Fig. 4 relate to the TEM-Researcher software, and the right panels relate to IX1D software; (a) and (b) present transient model curves, and (c) and (d) show inverse resistivity versus depth functions, respectively. The numbers denote the following:

(1) Conventional apparent resistivity versus time curve $\rho_a(t)$, which is an apparent resistivity of a homogenous medium the response of which at a given moment coincides with the signal measured

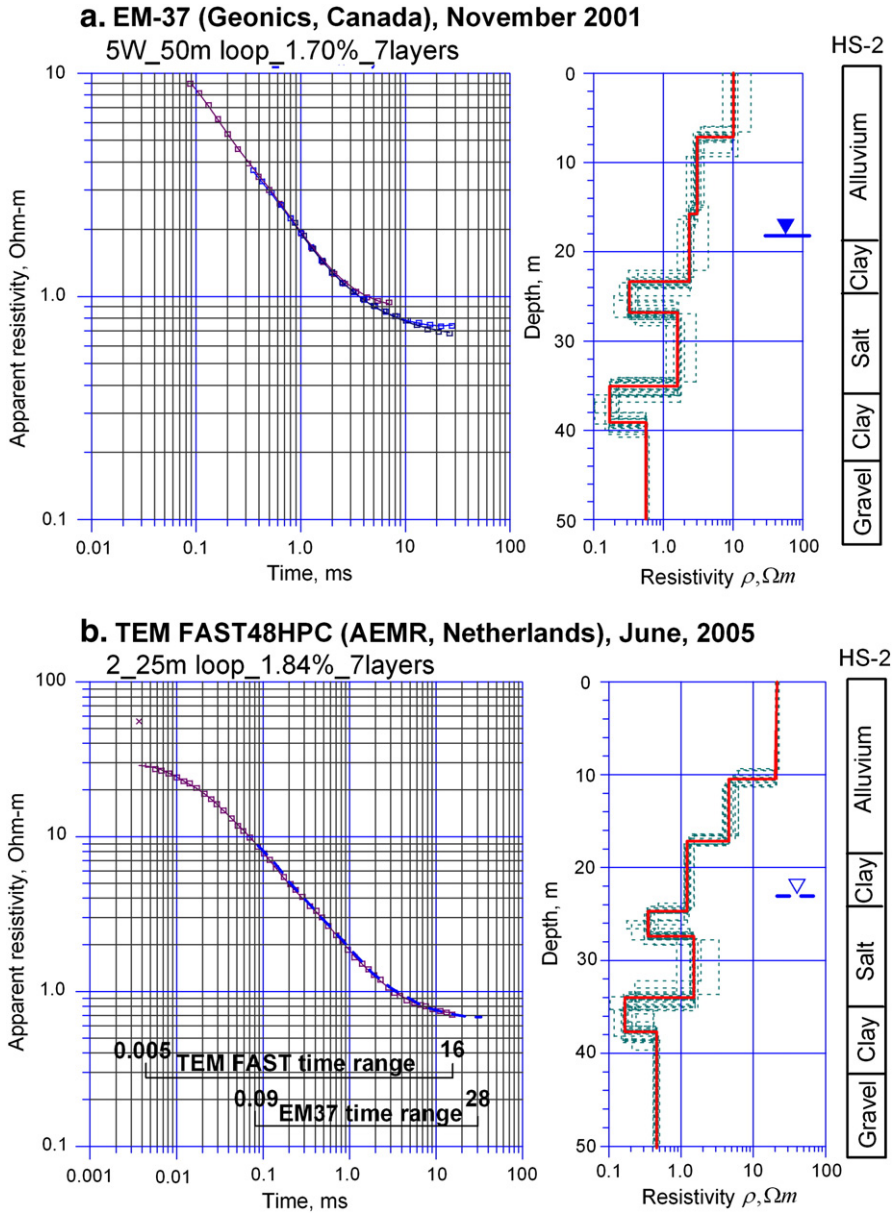


Fig. 3. Transient curves (left) and inverse resistivity-depth functions (right) measured at the TEM 2 station using the EM-37 system in 2001 (a) and the TEM FAST 48HPC system in 2005 (b). Geological column of the HS-2(1) borehole is shown to the right of the graphs. Dotted line in Fig. 3b (left) shows transient curve of the Geonics EM37 (for comparison).

in the experiment. It is calculated by the asymptotic formula for late times in the near zone of the transient field (Eq. (A-1) in Appendix 1 after Barsukov et al. (2007)).

(2) The intermediate function $\rho_f(t)$ (in contrast to $\rho_a(t)$) is introduced as an apparent resistivity valid at any stage of the transient process. This is why curves 1 and 2 in Fig. 4a coincide at late times and differ at earlier times (Eq. (A-2) in Appendix 1).

(3) Transformation curve $\rho_a(h)$ calculated from the function $\rho_f(t)$ and its derivative $\rho_f'(t)$ in the TEM FAST methodology.

(4) Inverse resistivity versus depth function is calculated from the transformation function in the TEM FAST methodology (Barsukov et al., 2007) and by the conventional Ridge Regression method in the IX1D software (Inman, 1975).

(5) Resistivity versus depth functions, showing equivalent solutions for the same $\rho_a(t)$ function.

The example in Fig. 4 demonstrates good agreement between the results of the inversion of one data set by two programs based on different principles of interpretation within the framework of the selected multi-layered model.

4. Estimation of the groundwater salinity in the DS shore area

There are several possibilities for estimating groundwater salinity in the DS area using geoelectric methods: (a) direct calibration of the formation resistivity versus water salinity (Ezersky et al., 2008a; Ezersky and Legchenko, 2009; Yechieli et al., 2001); (b) joint use of TEM and magnetic resonance sounding (MRS) methods (Legchenko et al., 2009), and (c) qualitative analysis of the relationship between the formation resistivity and water salinity (Kafri et al., 1997).

Using TEM resistivity measurements in the DS area Kafri et al. (1997) have identified several conventional interfaces separating zones of different salinities (Fig. 5): (I) the western zone with an electrical resistivity of $\rho>3$ Ωm, identified as fresh (or brackish) water; (II) the intermediate zone identified as diluted brine with a resistivity of 1 Ωm$<\rho<$3 Ωm; and (III) the eastern zone nearest the DS with a resistivity of $\rho<1$Ωm identified as DS brine.

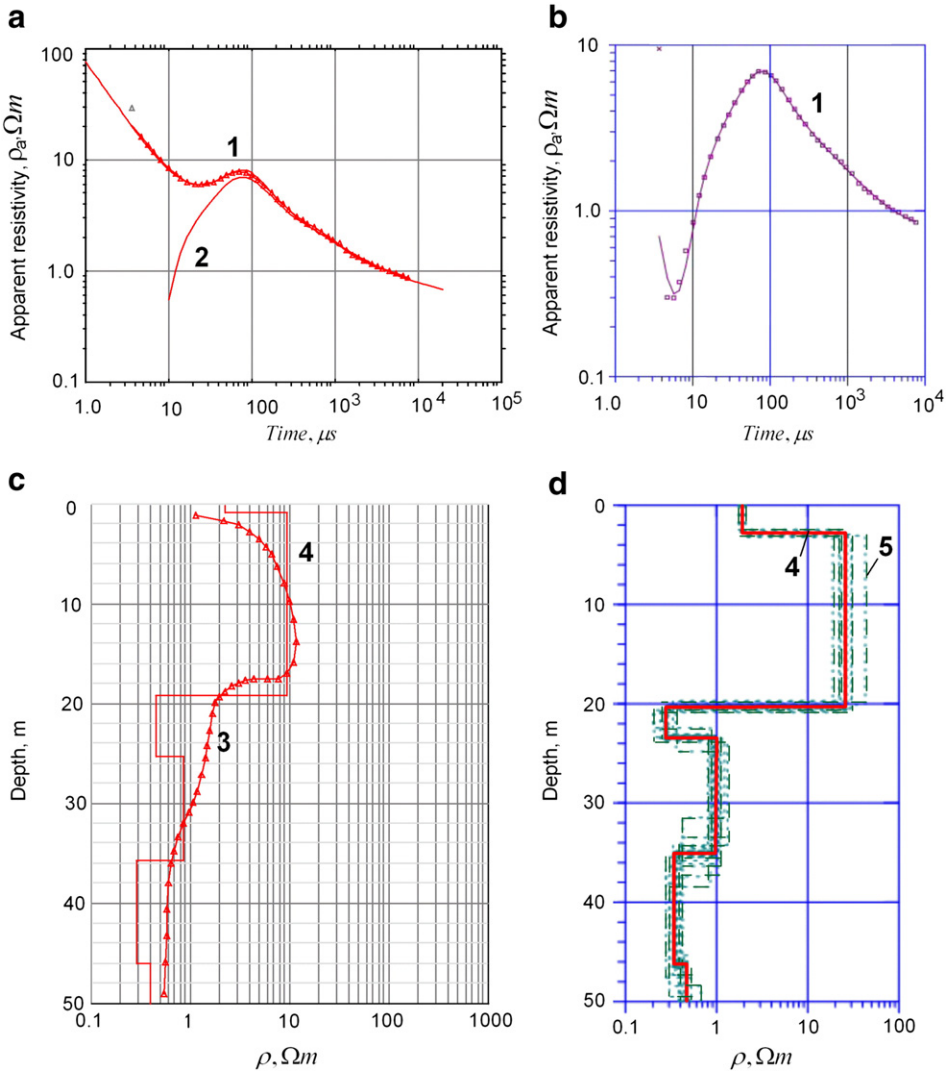


Fig. 4. Example of the interpretation of data set obtained at TEM station 21 (see Fig. 6 for location) using two processing software IX1D (by Interpex) and TEM-Researcher (AEMR, Netherlands), respectively: (a) and (b) — transient model curves and (c) and (d) — inverse resistivity versus depth functions. See text for numbers explanation.

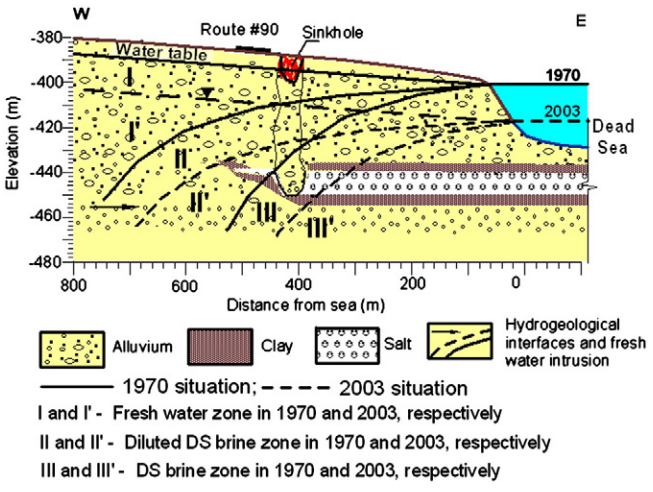


Fig. 5. Generalized geoelectrical section along the western shore of the DS. Solid lines show the interfaces between water quality zones in 1970 and dashed lines are from 2003.

The lowering of the Dead Sea level causes a corresponding lowering of the interfaces between the zones with different electrical resistivities (Yechieli et al., 2002). This may cause the salt edge to come into contact with less saline water.

Thus, the 1 Ωm interface between the diluted brine and the DS brine (in Fig. 5 — interface between zones II and III) is an important feature for estimating the potential of salt dissolution. Salt edge location is known from a seismic refraction survey (Ezersky et al., 2008b). This interface is either between unsaturated sediments or between fresh (brackish) water-saturated sediments from above and sediments saturated with the DS brine from below. Sometimes a 1 Ωm interface coincides with top of the clay layer keeping the DS brine for a long time after the lowering of the DS level.

The 1 Ωm interface could be characterized by two parameters: depth to the 1 Ωm interface (h_L), and the bulk resistivity below the interface (ρ_L). In surveyed areas the TEM soundings reliably define these parameters because of the sharp drop in the resistivity from several tens of Ωm to less than 1 Ωm.

Thus, the main objectives of our study were:

1. Mapping resistivity throughout the sinkhole development areas to identify less-saline water zones, especially in close proximity to the salt formation.

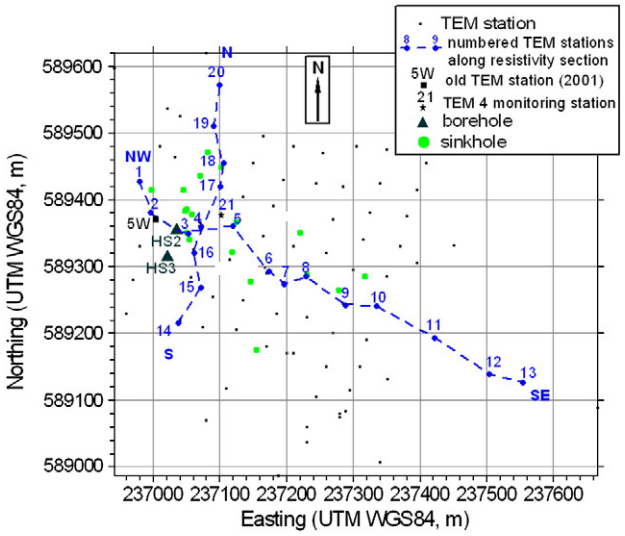


Fig. 6. A general location map showing all of the TEM stations, boreholes and sinkholes in 2005.

2. Locating the upper surface of the DS brine given by the depth of the 1 Ωm interface.

5. Results: geoelectric structure of the area

In 2005, 88 TEM soundings were made in an area of 600×600 m^2 in Nahal Hever South using 25×25 m^2 and 50×50 m^2 loops. The location map of these TEM stations is shown in Fig. 6.

a) Comparison of TEM results with HS-2 borehole log
In Fig. 7a and b TEM results are compared with the geological section of borehole HS-2 located close to TEM station 4.
The geological column in Fig. 7c is composed of five lithological units. A priori, we can expect different levels of resistivity of the selected units. Adding the factor of high salinity groundwater, we can expect considerable differentiation of the section with respect to resistivity. Indeed, uppermost alluvium has the resistivity of more than 10 Ωm, whereas underlain soil saturated with salty water (if it exists), and especially with DS brine, should have a very low resistivity. In water-saturated sediments, we select two layers of water-saturated clay enveloping the salt layer that, based on our preliminary knowledge derived from the laboratory measurements, has a resistivity of up to 5–9 Ωm. Finally, the layer underlying the lower clay unit comprises clay with gravel saturated with the DS brine. This unit is also expected to have a resistivity somewhat higher than that of saturated clay. Taking into account that layers of a higher and lower resistivity are interbedded, we can expect a priori at least six layers: unsaturated alluvium, saturated alluvium, saturated clay, salt, saturated clay, and a saturated sandy–gravel–clayey lowermost unit.

b) Resolution of the TEM method at the studied area
Despite the fact that more layers provide for a better inversion (Zhdanov and Keller, 1994), we do not want to add extra layers. We have to select the optimal model that, on one hand, correlates well with the geological section, and on the other provides the highest resolution of the layer parameters (resistivity ρ_i and layer thicknessh_i). The option to select a multi-layered model is a matter of equivalency and resolution. This issue has been discussed extensively in the literature (Fitterman et al., 1988; Goldman et al., 1994; Raiche et al., 1985; Sandberg, 1993). Stoyer has suggested a method for the estimation of equivalency and resolution based on Resolution Matrix Parameters (RMP) and implemented it in the IX1D software (Interpex Ltd., 2006; Stroyer, 2010). In the present paper, the brief analysis of the optimal multi-layered model is carried out to estimate the number of layers and the resolution of layer parameters. The optimal number of layers is selected based on the minimal misfit error (MRS), calculated according to Eq. (4). For instance, the resolution of the layer parameters for TEM station 4 (closest to borehole HS-2) is estimated using the Resolution Matrix. The experimental measurement points and the model transient curve are shown in Fig. 8a. Inverse resistivity versus depth functions is presented in Fig. 8b–f. The number of layers increases from left to right. The 6-layered model is presented in two versions: (1) the inversion was carried out with all parameters free (b–d, f); (2) data were inverted with a fixed layer thickness 5 (e). As the number of layers increases from 4 to 7 from left (b) to right (f), the RMS decreases. For the inversion procedure with all

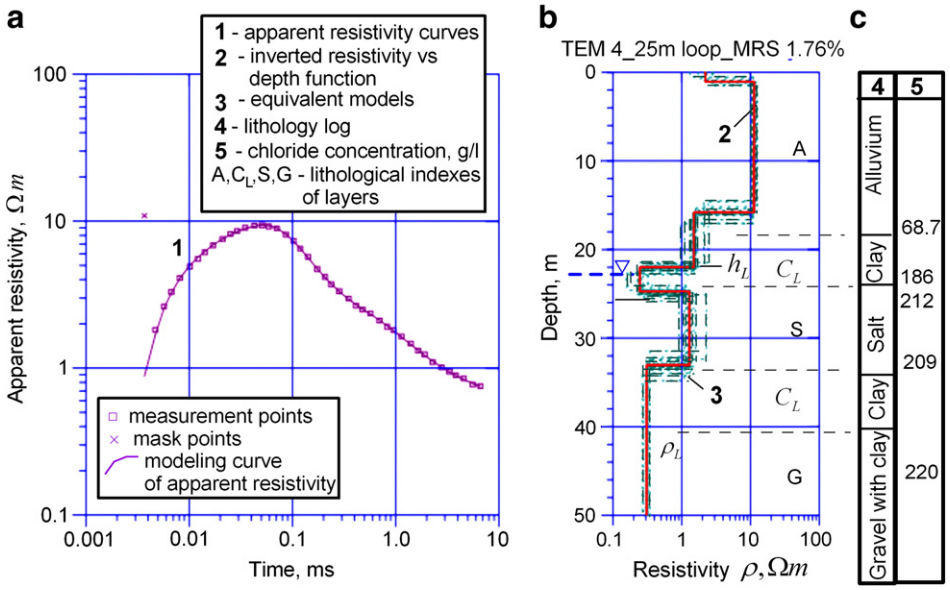


Fig. 7. TEM results from sounding 11–10 near HS-2 borehole: (a) TEM apparent resistivity versus time, and (b) inverted resistivity–depth function, (c) Hydrogeological column from borehole HS-2(1) (see Fig. 4 for location). Lithological indexes of the layers are designated by letters as A — alluvium above water table, C_L — clay saturated with the DS brine, S — salt, G — clay with gravel, saturated with the DS brine. Also noted is: h_L — depth from surface to 1 Ωm interface, and ρ_L — bulk resistivity of sediments saturated with the DS brine.

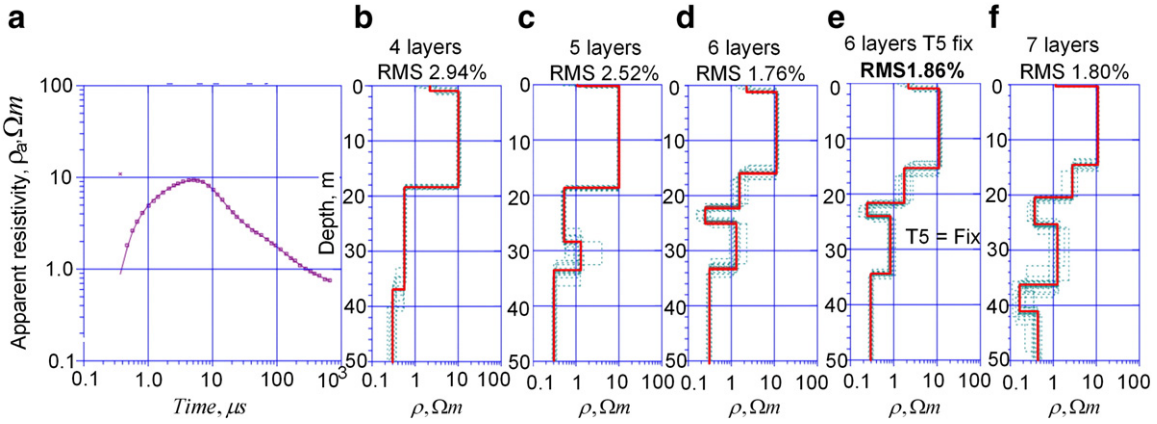


Fig. 8. (a) Experimental data (squares) and model transient curve (solid line) and (b–f) inverse resistivity versus depth functions calculated for different layered models: 4-layered (b), 5-layered (c), 6-layered (d and e), 7 layered (f). For models (b–d) and (f) all parameters are free, for model (e) thickness of the 5th layer (salt) was fixed, 3-layered model is not shown.

parameters free, an RMS error of 1.76 is minimal for the 6-layered model. It increases to 2.52% and 2.94% for the 5-layered and 4-layered models presented in Fig. 8c and b, respectively. For the 3-layered model, the RMS error achieves 7.07% (not shown in Fig. 8). It increases slightly to 1.80% for the 7-layered model as well (Fig. 8f).

The parameter bounds and the Resolution Matrix calculated for the model presented in Fig. 7 are presented in Tables 1 and 2.

As can be seen in Table 1, layer parameters vary in the range of 9–50%, generally showing the good inversion results. Maximum variations are characteristic for the resistivity of the 3rd and 5th layers and for the thickness of the first and fourth layers.

In Table 2 diagonal parameters and values higher than 0.3 are marked in bold. From the Resolution Matrix (Table 2) it can be determined that the conductance of the 1st layer is well constrained, the resistivity of the 2nd layer is well constrained, the thickness of the 2nd layer is well resolved, the resistivity of the 3rd layer is not well constrained, the thickness of the 3rd layer is fairly well constrained, the conductance of the 4th layer is well constrained, the resistivity of the 5th layer is not well constrained, the thickness of the 5th layer is fairly well constrained (there could be a slight transverse resistance equivalency), and the resistivity of the 6th (aquifer) layer is well constrained. Adopting the 7-layered model does not improve the RMS misfit error, but the resolution of parameters decreases, as can be seen in Fig. 8f. Both lower layers are characterized by high equivalency.

Contrary, accepting the 6-layered model we observe a good correlation between the resistivity versus depth function and geological column of the borehole (Fig. 7b and c). The equivalence is rather small and does not influence the results. Layer A located above the water table with resistivities of 1–20 Ωm corresponds to unsaturated alluvial sediments. This layer is composed mainly of unsaturated alluvium.

The A-layer is underlain by the clay layer (C_L), the upper part of which maintains a low resistivity (~1.5 Ωm), whereas the lower part is characterized by low resistivity of 0.3–0.4 Ωm. In other words, the clay layer is divided by a 1 Ωm interface at a depth of 23 m. The origin of the 1 Ωm interface has been discussed by Kafri et al. (1997).

In sandy aquifers, the 1 Ωminterface allows locating the upper boundary of sediments saturated with the DS brine (Goldman et al., 1991), and we expect that the 1 Ωm interface should coincide with the water table. However, in clay saturated with the DS brine this interface either does not follow the water table or follows it at a rate essentially lower than that in sands depending on the intensity of circulation of meteoric water (Yechieli et al., 2001). It seems to be the case. In 2001 the static water level was measured in borehole HS-2 at 17–18 m (−410 m altitude), and the clay was covered by brine. In 2003 HS-2 borehole collapsed. Therefore we suspect that in 2005 the water level dropped to a depth of 22–23 m (−415 m altitude) following the drop in the DS level to −419 m altitude. Accordingly, we suggest that the 1 Ωm interface measured in 2005 correlates with the DS brine groundwater table. Clay above the water table has been unsaturated with the DS brine.

Therefore depth to the low resistive layer h_L (corresponding to the 1 Ωm interface (in Fig. 7b)) correlates well to the water table in Fig. 7c.

The layer that appears at a depth of 26–33 m with a resistivity of 1.5 Ωm is identified as the salt unit (S), whereas low resistivity layers (0.2–0.4 Ωm) below are likely brine-saturated clay (C_L). We also note that the lowermost layer (G) does not exhibit a higher resistivity in comparison with the clay. This probably happens because gravel in this layer is mixed with a significant quantity of clay, and both are saturated with DS brine.

Considering the resolution of the most important layers, which are 5th (salt), and the 4th and 6th (saturated clay enveloping the salt): the 4th layer is characterized by longitudinal conductance equivalence (thickness–resistivity quotient), and is resolved well. But neither resistivity nor thicknesses alone are constrained. The resistivity of the 5th (salt) layer is not well constrained, whereas salt layer thickness is constrained well (RMP = 0.76). Finally, the resistivity of the lowest layer (RMP = 0.86) is constrained well. To improve the resolution of resistivity, we fixed the thickness of the 4th and 5th layers based on additional information from HS-2 borehole data (Fitterman et al., 1988; Sandberg, 1993). Constraining the

Table 1
Parameter bounds from equivalence analysis (Ωm). (All parameters are free).

	Min	Best	Max	Std%
Layer no				
1	1.32	2.22	2.86	±35
2	10.46	11.35	12.56	±9.4
3	0.98	1.52	2.45	±50
4	0.16	0.25	0.32	±32
5	0.91	1.3	2.28	±57
6	0.27	0.31	0.34	±11
Thickness				
1	0.52	1.03	1.54	±49.5
2	13.46	14.76	16.11	±9
3	5.17	6.23	7.3	±17
4	1.37	2.75	4.14	±50.4
5	6.64	8.3	10.09	±20.8

Table 2
Resolution Matrix Parameters (RMP), 6-layered model, misfit error 1.76%, All parameters are free.

Layer parameters	R1	R2	R3	R4	R5	R6	T1	T2	T3	T4	T5
Log(Ro1)	0.74	−0.02	0.02	0	0	0	−0.34	0.02	0.01	0	0
Log(Ro2)	−0.02	0.91	−0.04	0	0.00	0	−0.11	0.03	−0.08	−0.01	0
Log(Ro3)	0.02	−0.04	0.16	0.01	0.03	−0.03	0	0.16	−0.2	−0.07	−0.03
Log(Ro4)	0	0	0.01	0.58	0.12	−0.01	0	−0.01	0.14	−0.42	0.03
Log(Ro5)	0	0	0.03	0.12	0.10	0	0	0	−0.04	−0.1	0.22
Log(Ro6)	0	0	−0.03	−0.01	0.00	0.86	0.01	0	0.02	0.05	0.12
Log(T1)	−0.34	−0.11	0	0	0.00	0.01	0.46	0.05	−0.04	−0.01	0
Log(T2)	0.02	0.03	0.16	−0.01	0.00	0	0.05	0.96	0.08	0.02	0.01
Log(T3)	0.01	−0.08	−0.2	0.14	−0.04	0.02	−0.04	0.08	0.75	0.05	−0.01
Log(T4)	0	−0.01	−0.07	−0.42	−0.10	0.05	−0.01	0.02	0.05	0.36	0.02
Log(T5)	0	0	−0.03	0.03	0.22	0.12	0	0.01	−0.01	0.02	0.76

thickness of the 4th layer to 4 m and of the 5th layer to 11 m, we obtained resistivity resolutions of the 4th, 5th, and 6th layers of 0.84, 0.86, and 0.88, respectively. Note that the resolution of the 6th layer, as well as its equivalence, is better in the case of the 6-layered model than in the case of the 7-layered model.

c) Examples of TEM results obtained throughout the NHS area
Examples of the TEM results observed throughout the area along the NW–SE line are shown in Fig. 9. Fig. 9a presents typical responses of the apparent resistivity versus time. Corresponding inverse resistivity–depth functions are shown in Fig. 9b–h. To estimate the ambiguity of inverse depth models presented here, we also present several equivalent solutions (thin dotted lines).

Most of resistivity–depth functions (Fig. 9b) are characterized by similar structure corresponding generally to 6-layered resistivity–depth models. TEM identified resistivity layers are shown by dashed lines between soundings and designated as A, C_L, C_U, S and G. The resistivity generally decreases with depth from 10 to 12 Ωm in the uppermost subsurface to less than 1 Ωm below 40 m (Fig. 7b).

A-layer. The thickness of the upper high-resistivity A-layer at the western part of the line (TEM stations 2–10 in Fig. 9b–g) is about

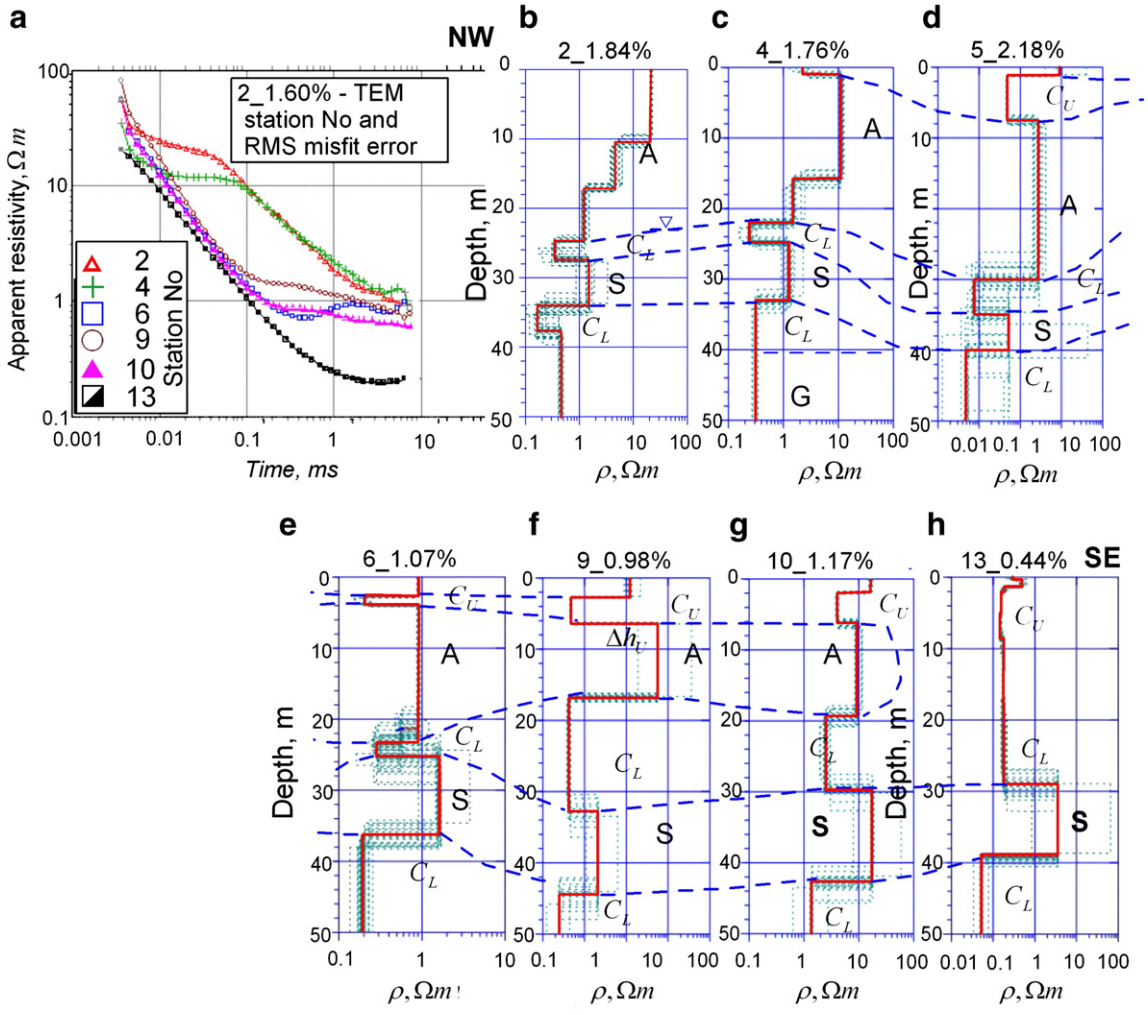


Fig. 9. Typical Nahal Hever TEM responses: (a) apparent resistivity versus time, and (b–h) corresponding inverse resistivity–depth (see Fig. 6 for location). Dashed lines connect layers designated by the lithological indexes (see Fig. 5 for explanation).

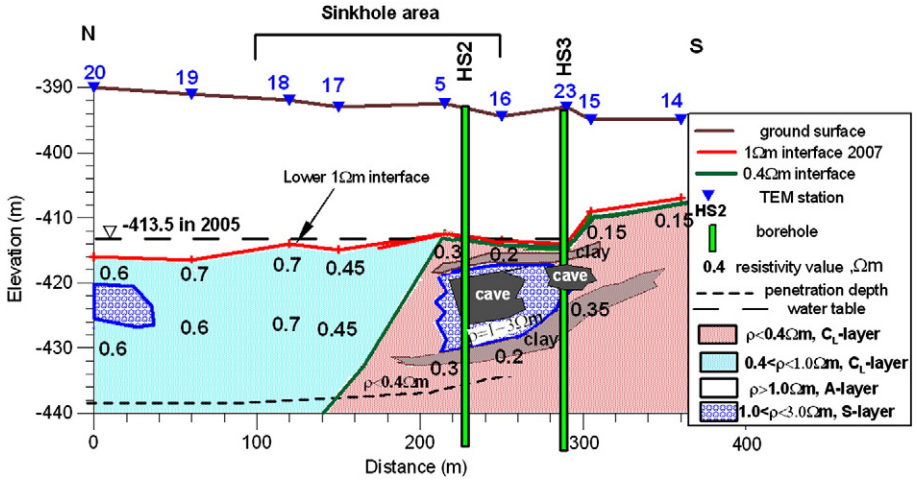


Fig. 10. Inverted resistivity cross-section along N–S Profile in 2005.

20–30 m. To the east of the area, this shallow high-resistivity layer transforms from 2.0 to 20 Ωm to a very low-resistivity layer of 0.2–0.3 Ωm (for instance, station 13 in Fig. 9h). This layer is composed of the clay keeping the DS brine relict after Dead Sea recession.

C_U-layer. At the central part of the section, near TEM station 6 (Fig. 9e), a new shallow layer with very low resistivity (<1.0 Ωm) appears, designated as C_U. It is clearly traced through stations 7–12 (Fig. 9e–g). This layer eventually disappears inside a thick layer with very low resistivity (0.2–0.3 Ωm), descending from the surface to a depth of 30 m (Fig. 9h). The layer is located above the water table. We suggest that it could be clay with a large amount of residual DS brine in the pore space. Such a layer could be caused by frequent historical fluctuations of the DS base level. The layer is not seen in boreholes HS-2 and HS-3 at the north-western TEM stations. However, a thin clay layer is clearly seen in the southern sinkholes, for example sinkhole #14 (see Fig. 13 for location).

C_L-layers. Layers C_L are also traced through entire section (Fig. 9b–g). Resistivity of the C_L-layer varies in the range of 0.2–0.3 Ωm. The thickness of this layer that is of 2.0–3.0 m in western TEM stations 2–6 (Fig. 9b–e) increases then at TEM stations 9–13 (Fig. 9f–h).

S-layer. Salt layer is traced also through station nos. 2–13 (Fig. 9b–h). The S-layer keeps the thickness at the range of 10–12 m and resistivity at the range of 1.5–3.0 Ωm.

C_L-layers. The underlying layer with a resistivity of less than 0.4 Ωm also suggests clay. This layer is underlain by gravel, with a significant clay content, and saturated with DS brine. Individual thin layers often cannot be resolved within this formation (C_L), because of the resolution limitations of the TEM method. The layer was traced throughout all the stations in Fig. 9.

Most of the inverse resistivity- versus depth functions presented in Fig. 9 are well-resolved and slightly equivalent. However, the depth functions of station 6, and especially station 5 (Fig. 9e and d, respectively) are characterized by a strong equivalency and bad resolution (especially at the lower part of the section). At first, the depth of the 1 Ωm interface is lowered at these stations to 23–30 m. The thickness of the salt layer decreases in Fig. 9d to 5 m.

d) Extension of the low resistivity layer (C_L).
In order to estimate the position of the salt layer relative to the 1 Ωm interface (h_L) we examined two resistivity cross-sections aligned in the N–S (profile 1) and NW–SE (profile 2) directions (see Fig. 6 for location).

TEM profile 1 (Fig. 10) crosses the main sinkhole agglomeration and shows that the resistivity in the southern part of the section is less than 0.4 Ωm indicating clay material. In clay, the TEM measurements cannot identify the DS brine level. A 1.5–3.0 Ωm-layer located within this clay formation corresponds to a salt layer (see also Fig. 8c). In the northern part however the resistivity varies between 0.45 and 0.7 Ωm what may correspond to sandy material saturated with DS brine (C_L layer shaded brown in Fig. 10). In sandy material the 1 *Ωm* interface shows the DS brine level. Thus, in 2005 when many sinkholes already existed salt was still surrounded by the DS brine.

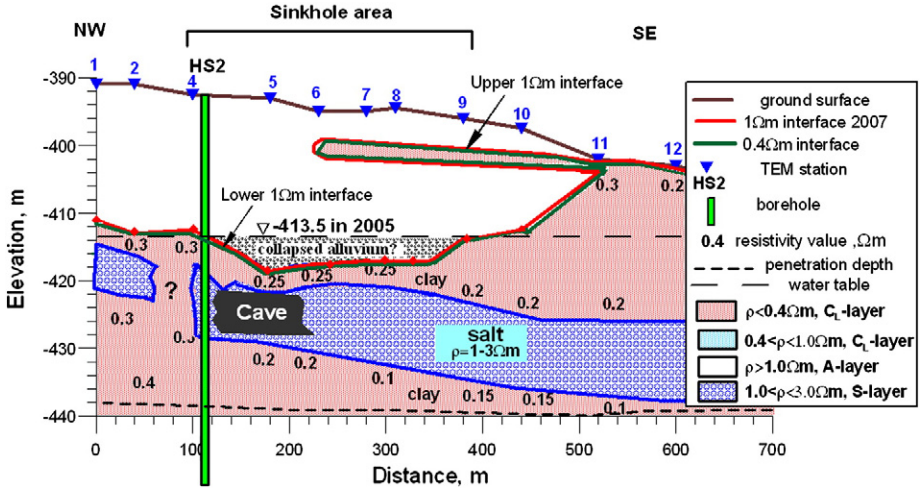


Fig. 11. Inverted resistivity section along NW–SE profile in 2005.

The resistivity cross-section along Profile 2 (NW–SE) shows the 1 Ωm interface disturbed by a subsidence above the salt layer (Fig. 11). This lowering is manifest in the elevation of the 1 Ωm interface (−417 to −418 m), which is approximately 5 m lower than the background elevation of the 1 Ωm interface throughout the area (e.g. −412 to −413 m). The Ωminterface is seen at a depth of 30 m at the resistivity–depth function of TEM station 5 (Fig. 9d) and at a depth of 23 m at station 6 (Fig. 9e). Ezersky et al. (2008b) have shown and discussed this depression in a 3D presentation. TEM station 5 is located close to the HS-2 borehole collapsing into the karstic cavity. We suggest that the strong equivalency and low resolution characteristics for stations 5 and 6 are associated with the cavity influence (3-D effect).

The salt layer is located within the sediments characterized by resistivity of 0.2–0.3 Ωm (C_L layer in Fig. 9 that is clay).

The TEM resistivity–depth section clearly shows a 0.2 Ωm layer under the salt layer that undoubtedly is caused by very high salinity ground water.

The spatial resistivity distribution of the sediments surrounding the salt layer (C_L layers in Fig. 9) is shown on the resistivity map through the NHS area (Fig. 12). One can see that resistivity of sediments in both maps (Fig. 12a and b) is <0.4 Ωm through central and southern parts of the area and increases slightly in the northern part (0.4–0.75 Ωm).

Thus, our results show that in 2005, when many sinkholes had appeared on the surface, salt was located within a very low resistivity environment inside sediments saturated with DS brine. We did not detect intrusion of relatively fresh water into the coastal aquifer.

e) Extension of the low resistivity layer (C_u).
The map of the thickness of the shallow low resistivity layer C_u in the NHS area is shown in Fig. 13. TEM results reveal a thin (1–5 m thick) layer interpreted as clay saturated with DS brine. This layer was observed only in the southern and south-eastern parts of the NHS area.

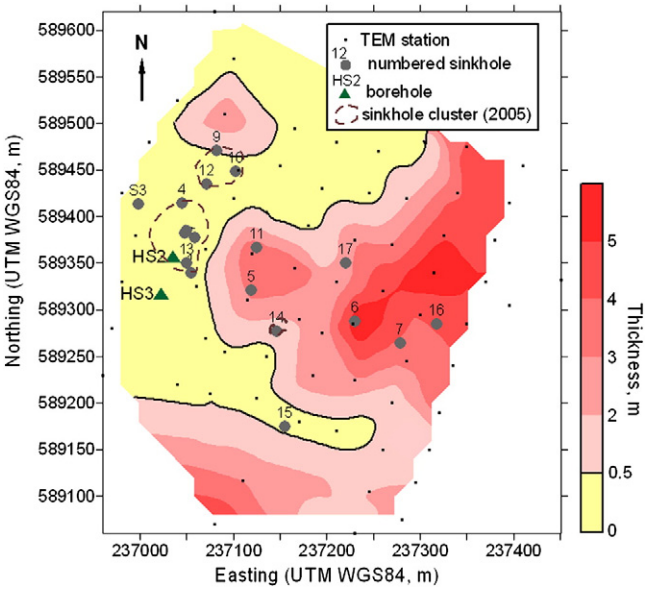


Fig. 13. Thickness map of the shallow, low resistive layer C_u in 2005.

6. Discussion

6.1. TEM results and 3D effects

Existence of sinkholes suggests that the subsurface may be rather heterogeneous. Consequently, 1D TEM measurements could be affected by 3D effects, which in turn may cause an erroneous interpretation of the TEM data. Rabinovich (1995) numerically studied the influence of the 3D effects on TEM interpretation, considering geology typical of Israel. Barsukov et al. (2007) investigated the influence of 3D heterogeneities on 1D TEM measurements using scale models made from aluminum foil and 3D forward modeling. The results of these studies have shown that in the presence of 3D electrical anomalies, 1D TEM measurements allow correct localization of the top of conductors and their horizontal extent. Thus, we estimate that even when 3D TEM interpretation is not used, 1D TEM results can provide a reasonable approximation of the subsurface resistivity. Obviously, in the vicinity of pronounced 3D structures, 3D interpretation, while much more complicated and expensive, would provide more accurate results.

6.2. Groundwater salinity

TEM measurements reveal that the resistivity of the subsurface varies in the range of 0.2–0.4 Ωm in the southern part of the Nahal

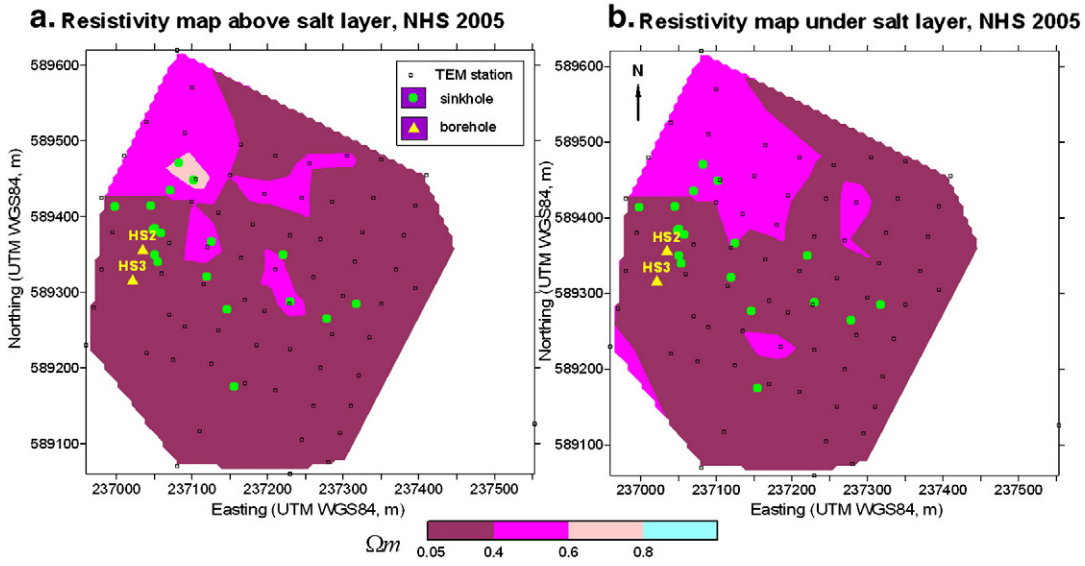


Fig. 12. Inverted resistivity map of the layers above (a) and below (b) the salt layer throughout the NHS area in June 2005.

Hever area, and resistivity increases to the northern one in the range of 0.6–0.7 Ωm (Fig. 12). Systematic TEM measurements between 2005 and 2008 (and some measurements carried out in 1999) revealed that over time, resistivity remained within the range of 0.3–0.6 Ωm thus confirming that since 1999 groundwater salinity was not changing (Ezersky et al., 2008b). These conclusions were recently confirmed by geochemical monitoring in boreholes (Yechieli, 2007).

Qualitative estimation of the water salinity using TEM measurements is based on the work of Kafri et al. (1997), who found that resistivity lower than 1 Ωm is an unambiguous indicator of DS brine in the pores of alluvial sediments. Our calibration of the resistivity versus salinity near 18 boreholes also confirms these results (Ezersky et al., 2008a; Ezersky and Legchenko, 2009). Results of the resistivity measurements near boreholes are summarized in Table 3.

It has been found that in the Nahal Hever area the typical value of the bulk resistivity in clay varies between 0.2 and 0.3 Ωm, whereas gravel and sandy sediments are characterized by a resistivity between 0.4 and 0.6 Ωm. The groundwater resistivity varies between 0.03 and 0.06 Ωm with corresponding chloride concentration in the range of 120–224 g/l (54% to 100% of saturation).

Knowing the groundwater resistivity, we could estimate the porosity using Eq. (2). Such estimation of the porosity of sandy aquifers is presented in Table 3. Our estimation reveals that the calculated porosity varies within a range of 23% to 31% with an average value of 27%. This agrees well with the results reported by Kafri and Goldman (2005) and Stephens et al. (1998) who estimated that the porosity of sandy–gravel sediments falls within the range of 26–43%.

Based on the calibration results we can conclude that the central and southern parts of the NHS area are composed of clayey lithology with resistivity less than 0.4 Ωm, whereas the northern part with resistivity of 0.45–0.7 Ωm is composed of sand and gravel all are saturated with highly saline water.

Based on the TEM study the salt layer in the NHS area is located within sediments saturated with DS brine of very high chloride concentration. Neither fresh nor brackish water was found either above or below the salt layer anywhere in a $600\times600\ m^2$ area investigated. Also there is no fresh or brackish groundwater to the west of the salt layer. Thus, the results discussed above are somewhat contradictory to the model that explains development of sinkholes through the fast dissolution of salt caused by intrusion of fresh groundwater into the aquifer. Sinkholes that appeared between 1997 and 2005 were formed throughout the NHS area at very high groundwater salinity (concentrated DS brine) enveloping the salt from all the sides. The TEM mapping did not reveal any structures characterized by the increased resistivity, which would suggest fresher water intrusion throughout the entire area, from above and from below.

6.3. Geoelectrical structure of the subsurface

We have shown that resistivity of the shallow subsurface reflects its vertical and lateral structures, i.e. its main hydrogeological elements explain the inter-relations between geology, hydrogeology, and sinkholes. The TEM method has allowed detailed differentiation of layers (clay, salt, etc.) in the subsurface based on their bulk resistivity. The 10 m-thick salt layer was extrapolated from borehole HS-2 throughout the study area. A very low resistive layer of the shallow part of the section can be considered as an example of the structural element of the subsurface.

In the southern and south-eastern parts of the study area the TEM results reliably reveal a thin (2–5 m thick) conductive layer composed of clay and silt that was saturated with DS brine only a few decades ago when the DS level was at an elevation of −390 m. After Yechieli et al. (2001) this layer is a DS brine relict; it could not have been flushed out of the clay and silt by meteoric water and lateral groundwater inflows because of the low hydraulic conductivity occurring at −400 m in the area close to the DS. The layer forms an aquiclude for recharge from above. Several shallow (2–3 m deep) sinkholes (#5, #11, #6, #7, #14, #15, #16, and #17) are located just above this clay layer suggesting that these sinkholes were not caused by the collapse of the ground into previously existing cavities within the salt layer (Ezersky et al., 2008b).

6.4. Recently developed sinkholes throughout the Nahal Hever South area

Results described in this paper relate to 2005. In the last 5 years, DS level was lowered by 5 m and the groundwater layer was lowered accordingly, reaching an elevation of approximately −418 m. Our last inspection of the area was performed in April 2010. We noted intensive sinkhole development somewhat north of the study area, as predicted by the MRS study (Legchenko et al., 2008b). However, the number of sinkholes throughout the study area did not increase, although some sinkholes increased their lateral sizes (#6, #7, #16, #17) and depth (#14). Our opinion is that the process of sinkhole development in the Nahal Hever South area reached a new stage. The saline groundwater level approached the salt and perhaps, in some places penetrated under it. Salt is protected from above and from below only by clay saturated with DS brine. When DS brine is washed out from clay pores, fresh water comes in contact with the salt top, dissolving it throughout the area. The model of sinkhole development along the salt edge suggested by Ezersky (2006), Ezersky et al. (2008b), and Legchenko et al. (2008a) needs to be changed for a late stage. Sinkholes will start to develop chaotically throughout the area and erosion will affect the surface. In this respect, the NHS area, characterized by the highest elevation of the salt layer can serve as a

Table 3
Bulk resistivity of the DS aquifer near calibration boreholes.

Borehole	Lithology	Chloride concentration measured in borehole g/l	Ground-water resistivity in borehole ρ_w, Ωm	Depth of layer measured in borehole m	Bulk resistivity measured with TEM ρ_x, Ωm,	Porosity estimated with TEM φ%,
Deragot-3	Gravel + sand	151.8	0.048[1]	30–50	0.6	25.5
Deragot-4	Gravel	200	0.047[1]	30–50	0.5	27.6
Mineral-1	Clay	120	0.055[2]	20–35	0.3	–
Mineral-2	Clay	205	0.046[2]	10–20	0.2	–
Mineral-4	Clay	216	0.05[2]	23–33	0.25	–
Ein-Boqeq2	Sand	230	0.045[a]	25	0.5	27
HS-2	Gravel + clay	208	0.047[a]	40–55	0.3	–
NZ-3	Clay	211	0.034[b]	15–35	0.3	–
EG-18	Sand	101	0.055[b]	32–60	0.55	28.4
EG-17	Gravel	204	0.047[a]	29–40	0.5	28
EG-11	Sand	198	0.05[b]	15–25	0.45	30

[a] Conductivity of groundwater was calculated using relationship after Ezersky and Legchenko (2009), Ezersky et al. (2008a) for high chloride concentrations.
[b] Conductivity of groundwater was measured by GSI either directly in boreholes or in the water samples.

model of future sinkhole development at other sites of the Dead Sea shore in Israel and Jordan.

7. Conclusion

Transient electromagnetic method (TEM) was applied to the Dead Sea sinkhole problem to map the electrical resistivity of the subsurface through the Nahal Hever South area which is typical for the DS coast. TEM measurements have allowed detailed differentiation of layers (clay, salt etc) in the subsurface based on their bulk resistivity. Results of our study reliably reveal that the electrical resistivity of the aquifer in the vicinity of the salt unit and its western border does not exceed 1 Ωm. (In most cases, electrical resistivity was measured between 0.2 and 0.6 Ωm.) This proves that the pore space of the alluvial sediments is filled with highly mineralized DS brine. Intrusion of relatively fresh groundwater into the aquifer through the 600×600 m^{2}area affected by sinkholes has not been detected. It was found that in Nahal Hever the typical value of the bulk resistivity in clay varies between 0.2 and 0.3 Ωm, whereas gravel and sandy sediments are characterized by the resistivity between 0.4 and 0.6 Ωm.

Based on results of calibration we can reliably conclude that the central and southern parts of the NHS area are composed of a clayey lithology with resistivity less than 0.4 Ωm, whereas the northern part with resistivities of 0.45–0.7 Ωm is composed of sand. Both areas are saturated with highly saline water.

These conclusions are in agreement with the earlier magnetic resonance sounding (MRS) findings that have explained the insignificant spatial increase of resistivity in the NHS area by lithological changes in the subsurface rather than variation of the groundwater salinity (Legchenko et al., 2009).

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Appendix 1

A common practice in electric prospecting is to use, as interpretation parameter, an apparent resistivity $\rho_a(t)$ of a homogeneous medium, the response of which at a given moment coincides with the signal measured in experiment. Traditionally, $\rho_a(t)$ is calculated by asymptotic formula *for late times in near zone* of transient field when the condition $t/(\mu_0 R^2/\rho_a(t))>>1$ is satisfied:

$$\rho_a(t)=\left[\frac{\sqrt{\pi}}{20}\frac{\mu^{5/2}R^4}{t^{5/2}E(t)/I}\right]^{2/3}. \qquad (A-1)$$

Here t is a transient time, μ and $\mu_0=4\pi x10^{-7}$ are magnetic permeability of medium and vacuum, respectively, $E(t)/I$ is the measured value of normalized voltage at antenna terminals, and $R=L/\pi^{1/2}$ the effective radius of a single-turn square antenna with a side L.

In the TEM FAST methodology apparent resistivity is calculated from the complete formula for the field valid at any stage of the transient process. The voltage in a single-loop antennae for homogeneous half space equals to:

$$E(t)/I=\mu\pi R^2\int_0^\infty\frac{e^{-t/\tau}}{\tau}\left[\frac{1}{\sqrt{\pi t/\tau}}-e^{t/\tau}erfc\left(\sqrt{t/\tau}\right)\right]J_1^2(\lambda r)d\lambda,\quad \tau=\frac{1}{\rho}\frac{\mu}{\lambda^2}. \qquad (A-2)$$

Apparent resistivity $\rho_f(t)$ is obtained by solving Eq. (A-2) for ρ, on the left-hand side of which there is the measured voltage.

Apparent resistivity calculated from Eq. (A-1) from asymptotic formula for the near-zone and from complete formula coincides at late times and essentially differs at earlier times where $t/(\mu_0R^2/\rho_a(t))<1$ (see Fig. 4a in text).

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